11. Comparing Two Samples

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Lecture Topics

- In Chapter 9, we studied the hypothesis testing framework for data drawn from a single distribution. In Chapter 11, we will study testing methods for data from two distributions and examining their differences.
- See Chapter 11 in **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007) for details. Selected examples and text excerpts are reproduced from [MSDA] for educational use only under fair use.
- This course presents content mostly following the textbooks, but also provides substantial unique and advanced material.
- Most of nonparameteric and Bayesian topics and Section 11.4 are assigned as reading materials and will not be covered in this course.

Lecture Topics: Two Sample Data

- To evaluate a new drug's efficacy, simply comparing patients who took the drug to those who took nothing is insufficient. Patients may improve merely because they believe they are being treated (the *placebo effect*).
- Therefore, we must compare a treatment group to a control group receiving a visually identical, inactive placebo.
- This comparison typically follows one of two designs:
  1. **Two Independent Samples**: Patients are randomly assigned to either the treatment or placebo group. The two groups do not have any common individuals:

$$\text{Treatment: } X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f_X \quad \text{and Placebo:} \quad Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} f_Y. \tag{1}$$

  2. **Paired Samples**: All the individuals are observed under both conditions (e.g., a crossover trial where a patient takes the placebo, undergoes a washout period, and then takes the active drug):

$$(X_1, Y_1)^T, \ldots, (X_n, Y_n)^T \overset{\text{i.i.d.}}{\sim} f_{X,Y}. \tag{2}$$

Lecture Topics: Two Sample Data

- We typically test whether the treatment is better than the placebo. For example, under the assumption of normality for two independent samples:

$$\text{Treatment: } X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma_X^2) \quad \text{and Placebo:} \quad Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma_Y^2), \tag{3}$$

  we can formulate the hypotheses as:

$$H_0 : \mu_X = \mu_Y \quad \text{vs.} \quad H_1 : \mu_X > \mu_Y. \tag{4}$$

- When the data are paired, we instead analyze the within-subject differences $D_i = X_i - Y_i$. Assuming:

$$D_1, \ldots, D_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_D, \sigma_D^2), \tag{5}$$

  the hypotheses are naturally simplified to a single-sample test on the differences:

$$H_0 : \mu_D = 0 \quad \text{vs.} \quad H_1 : \mu_D > 0. \tag{6}$$

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4. Experimental Design (*Reading Material)

Methods based on the Normal Distribution

- First, we explore cases where the population variances are known to be equal.
- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, where the parameter vector $\theta = (\mu_X, \mu_Y, \sigma^2)^T$ is unknown, $n \geq 2$, and $m \geq 2$.
- Suppose we test the hypotheses:

$$H_0 : \mu_X = \mu_Y \quad \text{vs.} \quad H_1 : \mu_X \neq \mu_Y. \tag{7}$$

- Then, the GLR test statistic is given by:

$$-2 \log \Lambda(\boldsymbol{X}, \boldsymbol{Y}) = (n+m) \log \left(1 + \frac{T^2}{n+m-2} \right), \tag{8}$$

where T is the two-sample t-statistic:

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}}, \quad \text{and} \quad S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}. \tag{9}$$

Methods based on the Normal Distribution

- Under $H_0 : \mu_X = \mu_Y$, we have:

$$Z = \frac{\bar{X} - \bar{Y}}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim \mathcal{N}(0,1) \quad \text{and} \quad V = \frac{(n+m-2)S_p^2}{\sigma^2} \sim \chi^2(n+m-2), \quad (10)$$

  and $Z \perp\!\!\!\perp V$.
- Therefore, $T \sim t(n+m-2)$, and the GLR test of level α is given by:

$$d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y}) = \begin{cases} 1 & \text{if } |T| > t_{\alpha/2}(n+m-2) \\ 0 & \text{if } |T| \leq t_{\alpha/2}(n+m-2). \end{cases} \quad (11)$$

Methods based on the Normal Distribution

Exercise 1

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, where the parameter vector $\theta = (\mu_X, \mu_Y, \sigma^2)^T$ is unknown, $n \geq 2$, and $m \geq 2$. Suppose we test $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$

a) Show that the MLE on $\Omega = \Omega_0 \cup \Omega_1$, $\hat{\theta}^{\Omega}$, is given by:

$$\hat{\theta}^{\Omega} = \left(\bar{X}, \bar{Y}, \frac{(n+m-2)S_p^2}{n+m} \right)^T, \tag{12}$$

where $S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$.

Methods based on the Normal Distribution

Methods based on the Normal Distribution

b) Show that the MLE on Ω_0, $\hat{\theta}^{\Omega_0}$, is given by $\hat{\theta}^{\Omega_0} = \left(\hat{\mu}^{\Omega_0}, \hat{\mu}^{\Omega_0}, \hat{\sigma^2}^{\Omega_0}\right)^T$, where

$$\hat{\mu}^{\Omega_0} = \frac{n\bar{X} + m\bar{Y}}{n+m} \quad \text{and} \quad \hat{\sigma^2}^{\Omega_0} = \hat{\sigma^2}^{\Omega} + \frac{nm}{(n+m)^2}\left(\bar{X} - \bar{Y}\right)^2. \tag{13}$$

Methods based on the Normal Distribution

Methods based on the Normal Distribution

c) Show that the GLR test statistic is given by:

$$-2\log\Lambda(\boldsymbol{X},\boldsymbol{Y}) = (n+m)\log\left(1+\frac{T^2}{n+m-2}\right), \tag{14}$$

where T is the two-sample t-statistic:

$$T = \frac{\bar{X}-\bar{Y}}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}}. \tag{15}$$

Methods based on the Normal Distribution

Methods based on the Normal Distribution

d) Since $T \sim t(n+m-2)$ under $H_0 : \mu_X = \mu_Y$, the GLR test of α is given by $d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y}) = I\left(|T| > t_{\alpha/2}(n+m-2)\right)$. Let $\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta[d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y})]$ be the power function of the GLR test. Show that:

$$\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta\left[1 - \Phi\left(\frac{S_p}{\sigma} t_{\alpha/2}(n+m-2) - \frac{\mu_X - \mu_Y}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}}\right) + \Phi\left(-\frac{S_p}{\sigma} t_{\alpha/2}(n+m-2) - \frac{\mu_X - \mu_Y}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}}\right)\right]. \tag{16}$$

Methods based on the Normal Distribution

Methods based on the Normal Distribution

- In the previous exercise, under H_1,

$$T = \frac{Z + \delta}{\sqrt{V/(n+m-2)}}, \tag{17}$$

where

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}}, \quad V = \frac{(n+m-2)S_p^2}{\sigma^2}, \quad \text{and} \quad \delta = \frac{\mu_X - \mu_Y}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}}. \tag{18}$$

Since $Z \perp\!\!\!\perp V$, T follows the noncentral t-distribution with $(n+m-2)$ degrees of freedom and noncentrality parameter $\delta = \frac{\mu_X - \mu_Y}{\sigma\sqrt{\frac{1}{n}+\frac{1}{m}}}$.

Methods based on the Normal Distribution

- Now, we explore cases where the population variances may be unequal.
- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma_X^2)$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma_Y^2)$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, where the parameter vector $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2)^T$ is unknown, $n \geq 2$, and $m \geq 2$.
- Suppose we test the hypotheses:

$$H_0 : \sigma_X^2 = \sigma_Y^2 \quad \text{vs.} \quad H_1 : \sigma_X^2 \neq \sigma_Y^2. \tag{19}$$

- Then, the GLR test statistic is given by:

$$\begin{aligned} -2\log\Lambda(\boldsymbol{X}, \boldsymbol{Y}) = &- n\log\frac{(n-1)W}{(n-1)W + (m-1)} - m\log\frac{(m-1)}{(n-1)W + (m-1)} \\ &- (n+m)\log(n+m) + n\log n + m\log m, \end{aligned} \tag{20}$$

where $W = S_X^2 / S_Y^2$ is the two-sample F-statistic.

Methods based on the Normal Distribution

- We have:

$$V_1 = \frac{(n-1)S_X^2}{\sigma_X^2} \sim \chi^2(n-1), \quad V_2 = \frac{(m-1)S_Y^2}{\sigma_Y^2} \sim \chi^2(m-1), \tag{21}$$

  and $V_1 \perp\!\!\!\perp V_2$.
- Therefore, under $H_0 : \sigma_X^2 = \sigma_Y^2$, $W = S_X^2/S_Y^2 \sim F(n-1, m-1)$, and the GLR test of level α is given by:

$$d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y}) = \begin{cases} 1 & \text{if } W < c_L \quad \text{or} \quad W > c_U \\ 0 & \text{if } c_L \leq W \leq c_U, \end{cases} \tag{22}$$

  where c_L and c_U are constants satisfying the conditions presented in the next exercise.

Methods based on the Normal Distribution

Exercise 2

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma_X^2)$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma_Y^2)$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, where the parameter vector $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2)^T$ is unknown, $n \geq 2$, and $m \geq 2$. Suppose we test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 \neq \sigma_Y^2$.

a) Show that the MLE on $\Omega = \Omega_0 \cup \Omega_1$, $\hat{\theta}^{\Omega}$, is given by:

$$\hat{\theta}^{\Omega} = \left(\bar{X}, \bar{Y}, \frac{(n-1)S_X^2}{n}, \frac{(m-1)S_Y^2}{m} \right)^T . \tag{23}$$

Methods based on the Normal Distribution

Methods based on the Normal Distribution

b) Show that the MLE on Ω_0, $\hat{\theta}^{\Omega_0}$, is given by:

$$\hat{\theta}^{\Omega_0} = \left(\bar{X}, \bar{Y}, \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m}, \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m} \right)^T. \tag{24}$$

Methods based on the Normal Distribution

Methods based on the Normal Distribution

c) Show that, for any $0 < \lambda_0 \leq 1$,

$$\Lambda(\boldsymbol{X}, \boldsymbol{Y}) < \lambda_0 \quad \Longleftrightarrow \quad \frac{S_X^2}{S_Y^2} < c_L \quad \text{or} \quad \frac{S_X^2}{S_Y^2} > c_U, \tag{25}$$

where the constants c_L and c_U satisfy $c_L \leq c_U$ and for $c \in \{c_L, c_U\}$,

$$\begin{aligned} \log \lambda_0 = \frac{n}{2} \log \frac{(n-1)c}{(n-1)c + (m-1)} + \frac{m}{2} \log \frac{(m-1)}{(n-1)c + (m-1)} \\ + \frac{n+m}{2} \log(n+m) - \frac{n}{2} \log n - \frac{m}{2} \log m. \end{aligned} \tag{26}$$

Methods based on the Normal Distribution

Methods based on the Normal Distribution

d) Let $\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta[d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y})]$ be the power function of the GLR test. Let $W \sim F(n-1, m-1)$. Show that:

$$\pi_{d^{\text{GLR}}}(\theta) = 1 - F_W\left(\frac{\sigma_Y^2 c_U}{\sigma_X^2}\right) + F_W\left(\frac{\sigma_Y^2 c_L}{\sigma_X^2}\right) \tag{27}$$

where c_L and c_U satisfy $c_L \leq c_U$,

$$\begin{aligned} & n \log \frac{(n-1)c_L}{(n-1)c_L + (m-1)} + m \log \frac{m-1}{(n-1)c_L + (m-1)} \\ & = n \log \frac{(n-1)c_U}{(n-1)c_U + (m-1)} + m \log \frac{m-1}{(n-1)c_U + (m-1)}, \end{aligned} \tag{28}$$

and $1 - \alpha = F_W(c_U) - F_W(c_L)$.

Methods based on the Normal Distribution

Methods based on the Normal Distribution

- Note that the GLR test is different from a typical equal-tail F-test given by:

$$d^{\text{Equal-tail}}(\boldsymbol{X}, \boldsymbol{Y}) = I\left(W < F_{1-\alpha/2}(n-1, m-1) \quad \text{or} \quad W > F_{\alpha/2}(n-1, m-1)\right). \tag{29}$$

- It is known that the equal-tail F-test may be biased. In contrast, the GLR test is unbiased.
- When both n and m are sufficiently large, the GLR and equal-tail F-tests are asymptotically equivalent.

A Nonparametric Method: The Mann-Whitney Test

- Until now, we have studied tests utilizing Gaussian assumptions.
- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F_X$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} F_Y$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$. Suppose X_i and Y_j are continuous random variables, and we test:

$$H_0 : F_X = F_Y \quad \text{vs.} \quad H_1 : F_X \neq F_Y. \tag{30}$$

  Note that we do not have parameters to describe F_X or F_Y.
- In such nonparametric cases, it is challenging to apply previous notions such as the GLR test, UMP test, etc.
- However, we can relax H_1 and construct some consistent test frameworks. The *Mann-Whitney Test* (also widely known as the *Wilcoxon Rank Sum Test*) is a prime example.

A Nonparametric Method: The Mann-Whitney Test

- In this lecture, we introduce the Mann-Whitney U statistic and skip detailed proofs.
- We compute the Mann-Whitney U statistic, which counts the number of times an observation from X is greater than an observation from Y:

$$U(\boldsymbol{X}, \boldsymbol{Y}) = \sum_{i=1}^{n} \sum_{j=1}^{m} I(X_i > Y_j). \tag{31}$$

  Note that $I(X_i > Y_j)$ are dependent.
- Assuming continuous distributions, under $H_0 : F_X = F_Y$, symmetry implies $P_0(X_i > Y_j) = 1/2$. Thus, the expected value of the statistic is:

$$\mathrm{E}_0[U(\boldsymbol{X}, \boldsymbol{Y})] = \frac{nm}{2}. \tag{32}$$

- It is known that, the variance under H_0 is given by:

$$\mathrm{Var}_0[U(\boldsymbol{X}, \boldsymbol{Y})] = \frac{nm(n+m+1)}{12}. \tag{33}$$

A Nonparametric Method: The Mann-Whitney Test

- Since $I(X_i > Y_j)$ are dependent, the standard CLT is not directly applicable to $U(\boldsymbol{X}, \boldsymbol{Y})$. Based on the theory of U-statistics, under $H_0 : F_X = F_Y$, we have:

$$Z = \frac{U(\boldsymbol{X}, \boldsymbol{Y}) - \frac{nm}{2}}{\sqrt{\frac{nm(n+m+1)}{12}}} \xrightarrow{d} \mathcal{N}(0, 1). \tag{34}$$

- Based on this, we can construct an asymptotic level α test as follows:

$$d^{\mathrm{MW}}(\boldsymbol{X}, \boldsymbol{Y}) = I\left(|Z| > z_{\alpha/2}\right). \tag{35}$$

- It is known that d^{MW} is consistent, i.e., its power converges to 1 as both n and m increases, when H_1 is given by:

$$P_1(X > Y) \neq \frac{1}{2}. \tag{36}$$

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Methods based on the Normal Distribution

- In this section, we study test methods when we have paired data. In contrast to the case of two independent samples, the collected data within a pair are inherently dependent.
- Let $(X_1, Y_1)^T, \ldots, (X_n, Y_n)^T \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \begin{pmatrix}\sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2\end{pmatrix}\right)$, where $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)^T$ is unknown and $n \geq 2$. Suppose we test $H_0 : \mu_X = \mu_Y \quad \text{vs.} \quad H_1 : \mu_X \neq \mu_Y$.
- The GLR test statistic is given by:

$$-2\log\Lambda((X_1, Y_1)^T, \ldots, (X_n, Y_n)^T) = n\log\left(1 + \frac{n\left(\bar{D}\right)^2}{(n-1)S_D^2}\right), \tag{37}$$

where $\bar{D} = \frac{1}{n}\sum_{i=1}^n D_i$ and $S_D^2 = \frac{1}{n-1}\sum_{i=1}^n (D_i - \bar{D})^2$ with $D_i = X_i - Y_i$.

Methods based on the Normal Distribution

- Although the GLR test statistic can be directly derived, the derivation is quite complex and is therefore provided as an advanced practice problem.
- As in most textbooks, we simplify this to a one-sample mean test by defining the difference $D_i = X_i - Y_i$. Let $D_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_D, \sigma_D^2)$, where the parameter vector $\theta = (\mu_D, \sigma_D^2)^T$ is unknown and $n \geq 2$.[1] Suppose we test $H_0 : \mu_D = 0 \quad \text{vs.} \quad H_1 : \mu_D \neq 0$.
- Based on the results derived in Chapter 9, the GLR test is given by:

$$d^{\text{GLR}}(\boldsymbol{D}) = I\left(\frac{\sqrt{n}|\bar{D}|}{S_D} > t_{\alpha/2}(n-1)\right). \tag{38}$$

[1] Based on the previous formulation, $\mu_D = \mu_X - \mu_Y$ and $\sigma_D^2 = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$.

Methods based on the Normal Distribution

Exercise 3

Suppose we design an experiment to test $H_0 : \mu_X = \mu_Y$ vs $H_1 : \mu_X \neq \mu_Y$. For simplicity, assume ρ is known and variances are known and equal: $\sigma_X^2 = \sigma_Y^2 = \sigma^2$. Consider two experimental designs using a total of $2n$ measurements ($n \geq 2$):

- **Two Independent Samples:** Draw n independent subjects for X and n independent subjects for Y, i.e., $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$.
- **Paired Samples:** Draw n pairs

$$(X_1, Y_1)^T, \ldots, (X_n, Y_n)^T \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \begin{pmatrix}\sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2\end{pmatrix}\right).$$

The GLR tests are equal-tail Z-tests. Show that the power for paired samples is larger than that for two independent samples if and only if $\rho > 0$.

Methods based on the Normal Distribution

Practice Problem I

1. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, where the parameter vector $\theta = (\mu_X, \mu_Y, \sigma^2)^T$ is unknown, $n \geq 2$, and $m \geq 2$. Suppose we test $H_0 : \mu_X \leq \mu_Y$ vs. $H_1 : \mu_X > \mu_Y$. Show that the GLR test of level α is given by:

$$d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y}) = \begin{cases} 1 & \text{if } T > t_\alpha(n+m-2) \\ 0 & \text{if } T \leq t_\alpha(n+m-2), \end{cases} \tag{39}$$

where T is the two-sample t-statistic:

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}}, \quad \text{and} \quad S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}. \tag{40}$$

Practice Problem II

2. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, \sigma_X^2)$, $Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, \sigma_Y^2)$, and $\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y}$, where the parameter vector $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2)^T$ is unknown, $n \geq 2$, and $m \geq 2$. Suppose we test $H_0 : \sigma_X^2 \leq \sigma_Y^2$ vs. $H_1 : \sigma_X^2 > \sigma_Y^2$. Show that the GLR test of level α is given by:

$$d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y}) = \begin{cases} 1 & \text{if } W > F_\alpha(n-1, m-1) \\ 0 & \text{if } W \leq F_\alpha(n-1, m-1), \end{cases} \tag{41}$$

where W is the two-sample F-statistic:

$$W = \frac{S_X^2}{S_Y^2}. \tag{42}$$

Advanced Practice Problem I

1. Let $(X_1, Y_1)^T, \ldots, (X_n, Y_n)^T \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \begin{pmatrix}\sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2\end{pmatrix}\right)$, where $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)^T$ is unknown and $n \geq 2$. Suppose we test $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$. The MLE over $\Omega = \Omega_0 \cup \Omega_1$, $\hat{\theta}^{\Omega}$ is given by:

$$\hat{\theta}^{\Omega} = \left(\bar{X}, \bar{Y}, \frac{(n-1)S_X^2}{n}, \frac{(n-1)S_Y^2}{n}, R\right)^T, \tag{43}$$

where R denotes the sample correlation coefficient.

Advanced Practice Problem II

a) Show that the MLE over Ω_0, $\hat{\theta}^{\Omega_0}$ is given by:

$$\hat{\theta}^{\Omega_0} = \left(\hat{\mu}^{\Omega_0}, \hat{\mu}^{\Omega_0}, \frac{(n-1)S_X^2}{n} + (\bar{X} - \hat{\mu}^{\Omega_0})^2, \frac{(n-1)S_Y^2}{n} + (\bar{Y} - \hat{\mu}^{\Omega_0})^2, \hat{\rho}^{\Omega_0} \right)^T, \tag{44}$$

where $\hat{\mu}^{\Omega_0}$ and $\hat{\rho}^{\Omega_0}$ are defined as:

$$\hat{\mu}^{\Omega_0} = \frac{(S_Y^2 - RS_XS_Y)\bar{X} + (S_X^2 - RS_XS_Y)\bar{Y}}{S_X^2 + S_Y^2 - 2RS_XS_Y},$$

$$\hat{\rho}_0 = \frac{\frac{n-1}{n}RS_XS_Y + (\bar{X} - \hat{\mu}^{\Omega_0})(\bar{Y} - \hat{\mu}^{\Omega_0})}{\sqrt{\left[\frac{(n-1)S_X^2}{n} + (\bar{X} - \hat{\mu}^{\Omega_0})^2\right]\left[\frac{(n-1)S_Y^2}{n} + (\bar{Y} - \hat{\mu}^{\Omega_0})^2\right]}}.$$

Advanced Practice Problem III

b) Show that the GLR test statistic is given by:[2]

$$-2\log\Lambda = n\log\left(1 + \frac{n(\bar{X}-\bar{Y})^2}{(n-1)(S_X^2 + S_Y^2 - 2RS_XS_Y)}\right). \tag{45}$$

[2] Note that $\bar{D} = \bar{X} - \bar{Y}$ and $S_D^2 = S_X^2 + S_Y^2 - 2RS_XS_Y$.

References I

Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.